Memory Management

Outline

- Background
- Contiguous Memory Allocation
- Paging
- Structure of the Page Table
- Swapping

Objectives

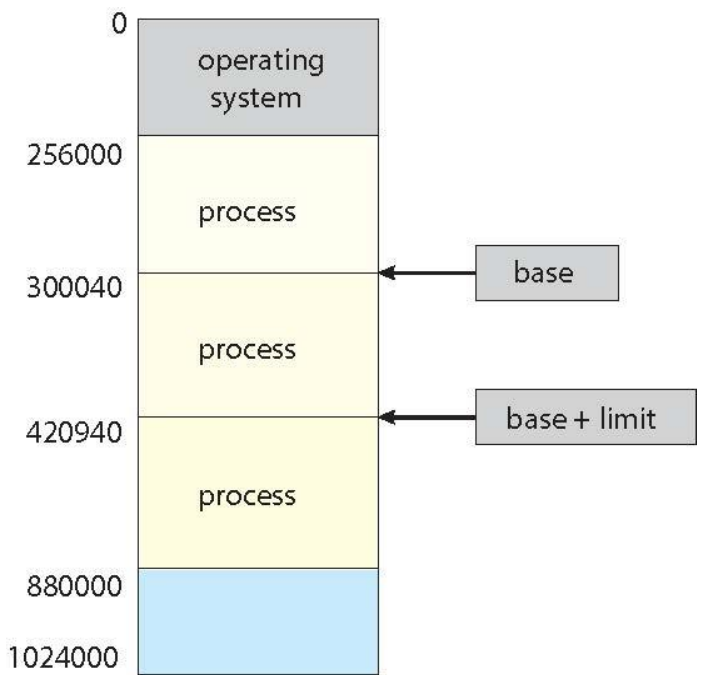
- To provide a detailed description of various ways of organizing memory hardware
- To discuss various memory-management techniques,
- To provide a detailed description of the Intel Pentium, which supports both pure segmentation and segmentation with paging

Background

- Program is permanently kept on **backing store** (disks)
- For a program to be run it must be brought from backing store into memory and placed within a process
- Main memory and registers are the only storage devices the CPU can access directly
- Memory unit only sees a stream of:
  - addresses + read requests, or
  - address + data and write requests
- Register access is done in one CPU clock (or less)
- Main memory access can take many cycles, causing a **stall**
- **Cache** sits between main memory and CPU registers
- Protection of memory is required to ensure correct operation

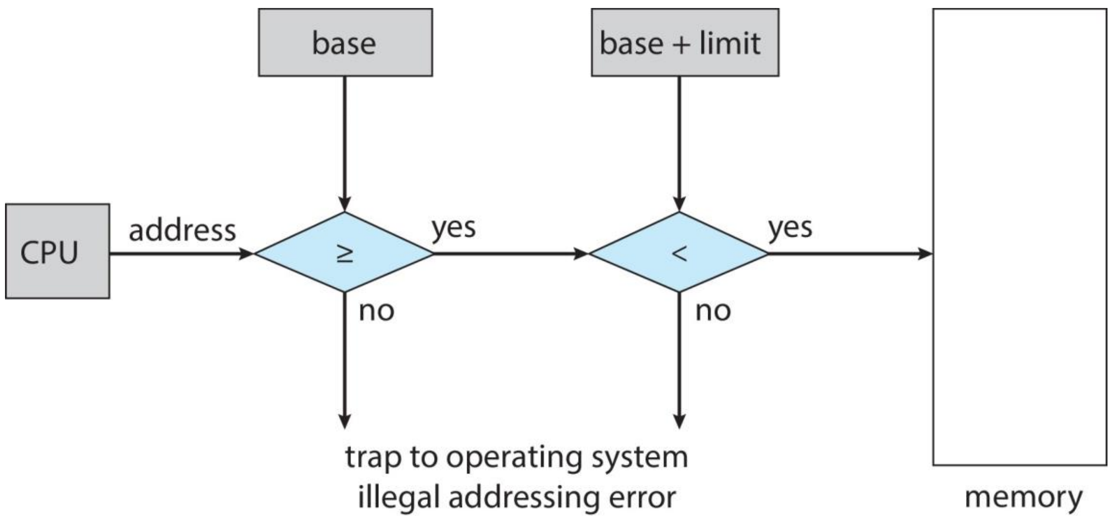
Memory Protection

- Need to censure that a process can access only access those addresses in it address space
- We can provide this protection by using a pair of **base** and **limit registers** define the logical address space of a process



Hardware Address Protection

- CPU must check every memory access generated in user mode to be sure it is between base and limit for that user



- The instructions to loading the base and limit registers are privileged

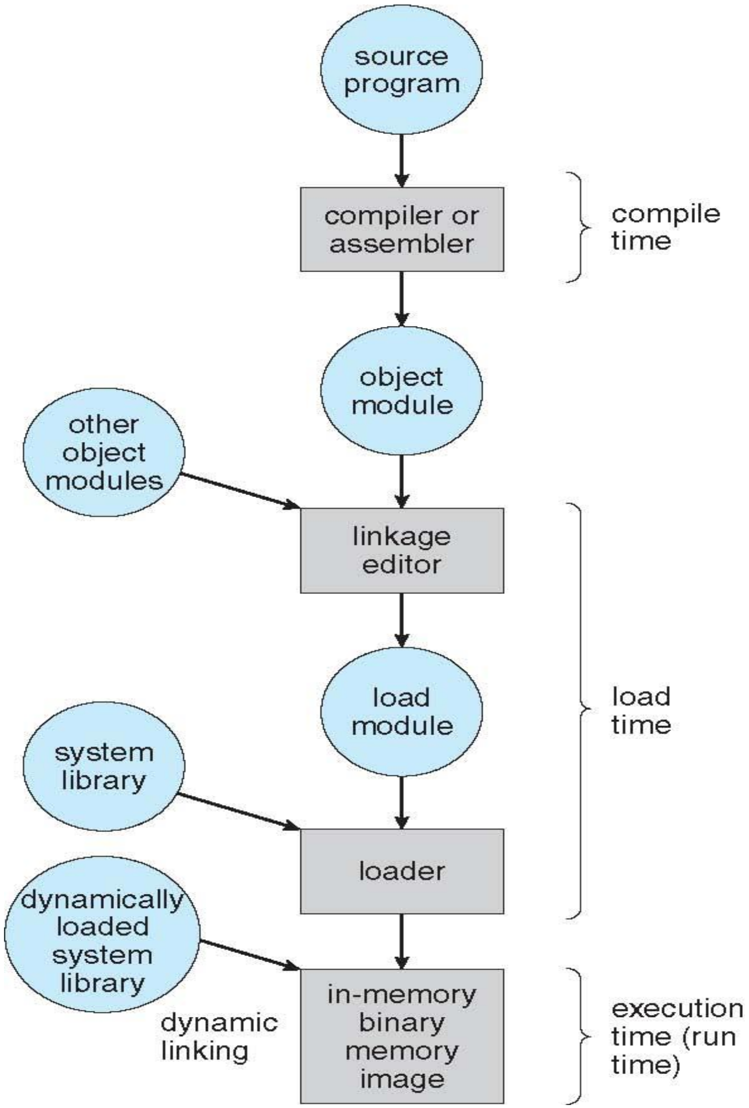
Address Binding

- Programs on disk, ready to be brought into memory to execute, are placed in an **input queue**
  - Without support, must be loaded into address 0000
- Inconvenient to have first user process physical address always at 0000
  - How can it not be?
- Addresses represented in different ways at different stages of a program's life
  - Source code addresses are usually symbolic
  - Compiled code addresses **bind** to relocatable addresses
    - i.e., "14 bytes from beginning of this module"
  - Linker or loader will bind relocatable addresses to absolute addresses
    - i.e., 74014
  - Each binding maps one address space to another

Binding of Instructions and Data to Memory

- Address binding of instructions and data to memory addresses can happen at three different stages
  - **Compile time**: If memory location known a priori, **absolute code** can be generated; must recompile code if starting location changes
  - **Load time**: Must generate **relocatable code** if memory location is not known at compile time
  - **Execution time**: Binding delayed until run time if the process can be moved during its execution from one memory segment to another
    - Need hardware support for address maps (e.g., base and limit registers)

Multistep Processing of a User Program

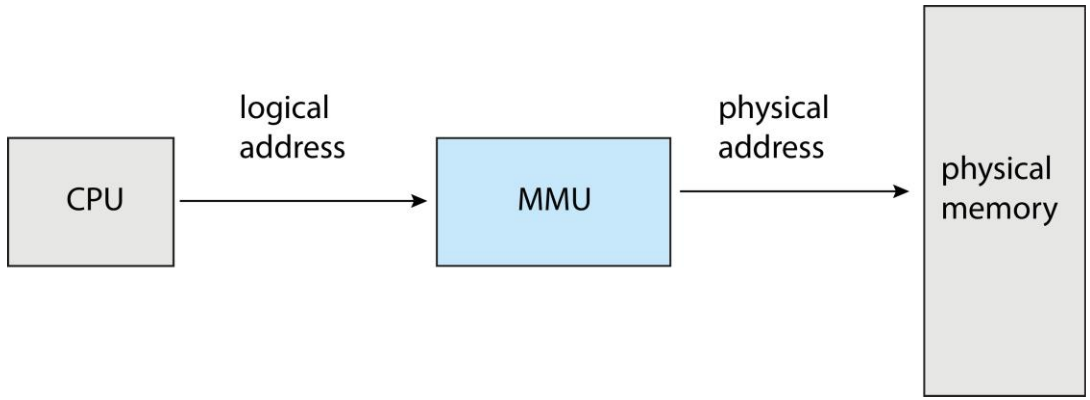


Logical vs. Physical Address Space

- The concept of a logical address space that is bound to a separate **physical address space** is central to proper memory management
  - **Logical address** – generated by the CPU; also referred to as **virtual address**
  - **Physical address** – address seen by the memory unit
- Logical and physical addresses are the same in compile-time and load-time address-binding schemes; logical (virtual) and physical addresses differ in execution-time address-binding scheme
- **Logical address space** is the set of all logical addresses generated by a program
- **Physical address space** is the set of all physical addresses generated by a program

Memory-Management Unit (MMU)

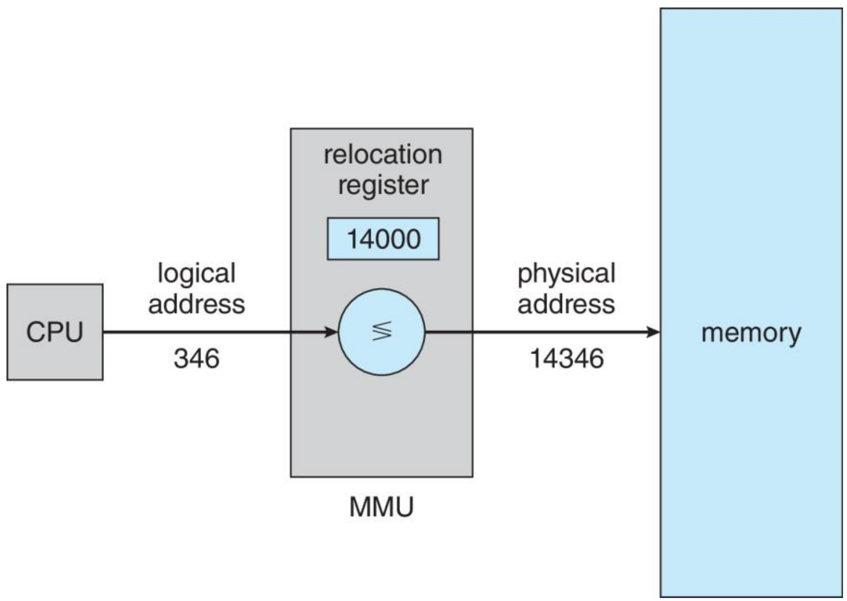
- Hardware device that at run time maps virtual to physical address



- Many methods are possible

MMU -- Relocation Register

- A simple scheme, which is a generalization of the base-register scheme.
- The base register now called **relocation register**
- The value in the relocation register is added to every address generated by a user process at the time it is sent to memory



Relocation Register (Cont.)

- The user program deals with *logical* addresses; it never sees the *real* physical addresses
  - Execution-time binding occurs when reference is made to location in memory
  - Logical address bound to physical addresses

Dynamic Loading

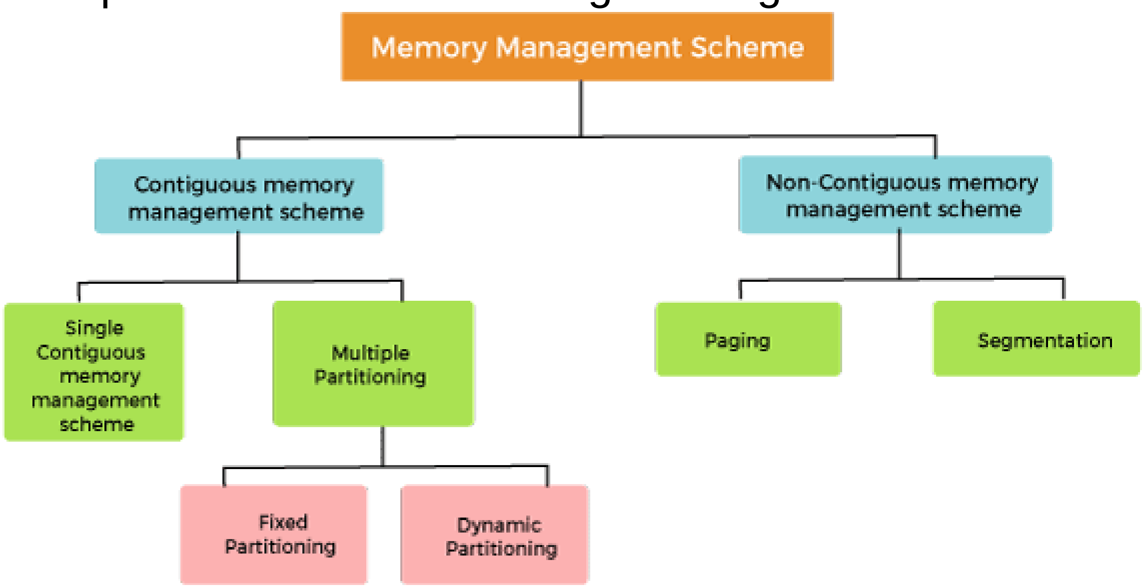
- The program consist of main part and a number of routines
- The entire program does need to be in memory to execute
- Routine is not loaded until it is called
- Better memory-space utilization; unused routine is never loaded
- All routines kept on disk in relocatable load format
- Useful when large amounts of code are needed to handle infrequently occurring cases
- No special support from the operating system is required
  - Implemented through program design
  - OS can help by providing libraries to implement dynamic loading

Dynamic Linking

- **Static linking** – system libraries and program code combined by the loader into the binary program image
- Dynamic linking – linking postponed until execution time
- Small piece of code, called **stub**, is used to locate the appropriate memory-resident library routine
- Stub replaces itself with the address of the routine, and executes the routine
- Operating system checks if routine is in processes' memory address
  - If not in address space, add to address space
- Dynamic linking is particularly useful for libraries
- System also known as **shared libraries**
- Consider applicability to patching system libraries
  - Versioning may be needed

Memory Allocation

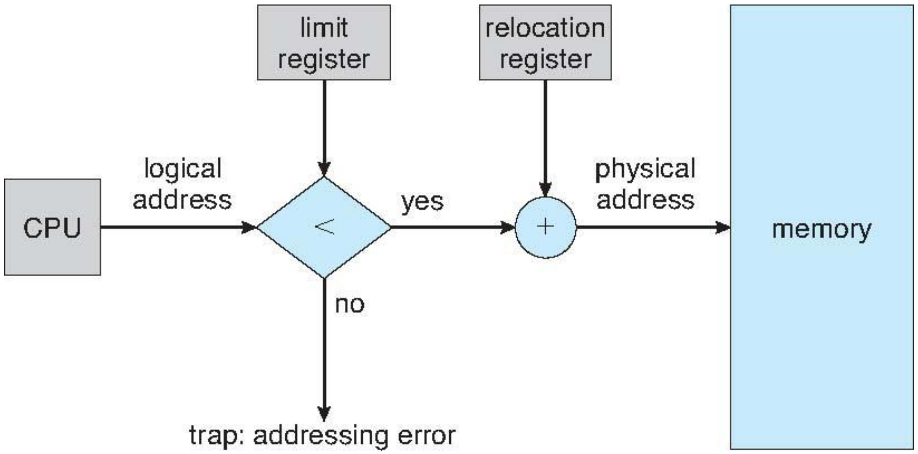
- Main memory must support both OS and user processes
- Limited resource, must allocate efficiently
- **Contiguous allocation** is one early method
  - Main memory usually consists of two **partitions**:
    - Resident operating system, usually held in low memory with interrupt vector
    - User processes then held in high memory
  - Each process contained in single contiguous section of memory



Classification of memory management schemes

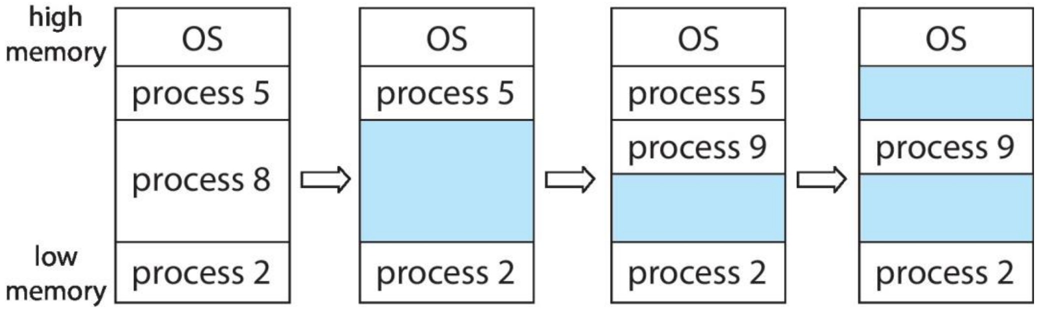
Memory Protection

- Relocation registers used to protect user processes from each other, and from changing operating-system code and data
  - Base register contains value of smallest physical address
  - Limit register contains range of logical addresses – each logical address must be less than the limit register
  - MMU maps logical address *dynamically*
  - Can then allow actions such as kernel code being **transient** and kernel changing size



Variable Partition Allocation

- Degree of multiprogramming limited by number of partitions
- **Variable-partition** sizes for efficiency (sized to a given process' needs)
- **Hole** – block of available memory; holes of various size are scattered throughout memory
- When a process arrives, it is allocated memory from a hole large enough to accommodate it
- Process exiting frees its partition, adjacent free partitions combined
- Operating system maintains information about:
  - (a) allocated partitions
  - (b) free partitions (holes)



Dynamic Storage-Allocation Problem

- How to satisfy a request of size ***n*** from a list of free holes?
  - **First-fit**: Allocate the ***first*** hole that is big enough
  - **Best-fit**: Allocate the ***smallest*** hole that is big enough; must search entire list, unless ordered by size
    - Produces the smallest leftover hole
  - **Worst-fit**: Allocate the ***largest*** hole; must also search entire list
    - Produces the largest leftover hole
- First-fit and best-fit better than worst-fit in terms of speed and storage utilization

First Fit Allocation in OS

90	50	30	40
Process 1	Process 2	Process 3	Process 4

20	100	40	200	10
Block 1	Block 2	Block 3	Block 4	Block 5
Available	Available	Available	Available	Available

90
Process 1

20	100	40	200	10
Block 1	Block 2	Block 3	Block 4	Block 5
Available	Available	Available	Available	Available

Block / Size	Available	Can Occupy P1?	Memory Wastage After Process Occupies
Block 1 (20K)	Yes	No	
Block 2 (100K)	Yes	Yes	100 - 90 = 10
Block 3 (40K)	Yes		
Block 4 (200K)	Yes		
Block 5 (10K)	Yes		

Block 2 is the first to Fit P1

P1 goes to Block 2

50
Process 2

20	100	40	200	10
Block 1	Block 2	Block 3	Block 4	Block 5
Available	Unavailable	Available	Available	Available

Block / Size	Available	Can Occupy P2?	Memory Wastage After Process Occupies
Block 1 (20K)	Yes	No	
Block 2 (100K)	No		
Block 3 (40K)	Yes	No	
Block 4 (200K)	Yes	Yes	200 - 50 = 150
Block 5 (10K)	Yes		

Block 4 is the first to Fit P2

P2 goes to Block 4

30
Process 3

20	100	40	200	10
Block 1	Block 2	Block 3	Block 4	Block 5
Available	Unavailable	Available	Unavailable	Available

Block / Size	Available	Can Occupy P3?	Memory Wastage After Process Occupies
Block 1 (20K)	Yes	No	
Block 2 (100K)	No		
Block 3 (40K)	Yes	Yes	40 - 30 = 10
Block 4 (200K)	No		
Block 5 (10K)	Yes		

Block 3 is the first to Fit P3

P3 goes to Block 3

40
Process 4

20	100	40	200	10
Block 1	Block 2	Block 3	Block 4	Block 5
Available	Unavailable	Unvailable	Unavailable	Available

Block / Size	Available	Can Occupy P3?	Memory Wastage After Process Occupies
Block 1 (20K)	Yes	No	
Block 2 (100K)	No		
Block 3 (40K)	No		
Block 4 (200K)	No		
Block 5 (10K)	Yes	No	

None of the available blocks
Can accommodate P4. It remains unallocated

Best Fit Allocation in OS

40	10	30	60
Process 1	Process 2	Process 3	Process 4

100	50	30	120	35
Block 1	Block 2	Block 3	Block 4	Block 5

40		100	50	30	120	35
Process 1		Block 1	Block 2	Block 3	Block 4	Block 5
		Available	Available	Available	Available	Available

Block / Size	Available	Can Occupy P1?	Memory Wastage After Process Occupies	
Block 1 (100K)	Yes	Yes	100 - 40 = 60	
Block 2 (50K)	Yes	Yes	50 - 40 = 10	Best
Block 3 (30K)	Yes	No	-	
Block 4 (120K)	Yes	Yes	120 - 40 = 80	
Block 5 (35K)	Yes	No	-	

Block 2 results in the least memory wastage

P1 goes to Block 2

10		100	50	30	120	35
Process 2		Block 1	Block 2	Block 3	Block 4	Block 5
		Available	Unavailable	Available	Available	Available

Block / Size	Available	Can Occupy P2?	Memory Wastage After Process Occupies	
Block 1 (100K)	Yes	Yes	100 - 10 = 90	
Block 2 (50K)	No	-	-	
Block 3 (30K)	Yes	Yes	30 - 10 = 20	Best
Block 4 (120K)	Yes	Yes	120 - 10 = 110	
Block 5 (35K)	Yes	Yes	35 - 10 = 25	

Block 3 results in the least memory wastage

P2 goes to Block 3

30		100	50	30	120	35
Process 3		Block 1	Block 2	Block 3	Block 4	Block 5
		Available	Unavailable	Unavailable	Available	Available

Block / Size	Available	Can Occupy P3?	Memory Wastage After Process Occupies	
Block 1 (100K)	Yes	Yes	100 - 30 = 70	
Block 2 (50K)	No	-	-	
Block 3 (30K)	Yes	-	-	
Block 4 (120K)	Yes	Yes	120 - 30 = 90	
Block 5 (35K)	Yes	Yes	35 - 30 = 5	Best

Block 5 results in the least memory wastage

P3 goes to Block 5

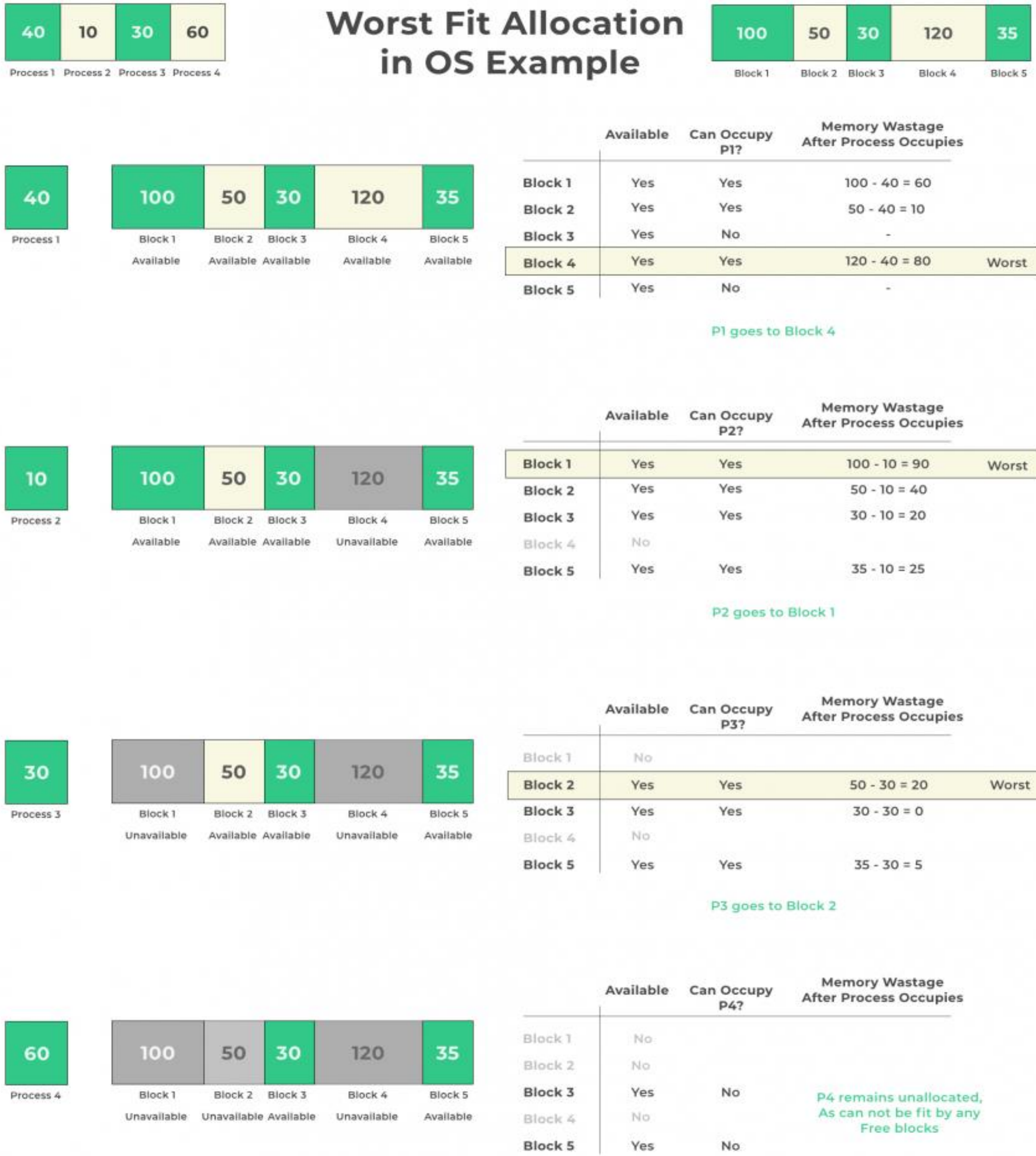
60		100	50	30	120	35
Process 4		Block 1	Block 2	Block 3	Block 4	Block 5
		Available	Unavailable	Unavailable	Available	Unavailable

Block / Size	Available	Can Occupy P3?	Memory Wastage After Process Occupies	
Block 1 (100K)	Yes	Yes	100 - 60 = 40	Best
Block 2 (50K)	No	-	-	
Block 3 (30K)	Yes	-	-	
Block 4 (120K)	Yes	Yes	120 - 60 = 60	
Block 5 (35K)	Yes	-	-	

Block 1 results in the least memory wastage

P4 goes to Block 1

Worst Fit Allocation in OS Example



	Available	Can Occupy P1?	Memory Wastage After Process Occupies	
Block 1	Yes	Yes	100 - 40 = 60	
Block 2	Yes	Yes	50 - 40 = 10	
Block 3	Yes	No	-	
Block 4	Yes	Yes	120 - 40 = 80	Worst
Block 5	Yes	No	-	

P1 goes to Block 4

	Available	Can Occupy P2?	Memory Wastage After Process Occupies	
Block 1	Yes	Yes	100 - 10 = 90	Worst
Block 2	Yes	Yes	50 - 10 = 40	
Block 3	Yes	Yes	30 - 10 = 20	
Block 4	No			
Block 5	Yes	Yes	35 - 10 = 25	

P2 goes to Block 1

	Available	Can Occupy P3?	Memory Wastage After Process Occupies	
Block 1	No			
Block 2	Yes	Yes	50 - 30 = 20	Worst
Block 3	Yes	Yes	30 - 30 = 0	
Block 4	No			
Block 5	Yes	Yes	35 - 30 = 5	

P3 goes to Block 2

	Available	Can Occupy P4?	Memory Wastage After Process Occupies
Block 1	No		
Block 2	No		
Block 3	Yes	No	
Block 4	No		
Block 5	Yes	No	

P4 remains unallocated, As can not be fit by any Free blocks

Example

- Given memory partitions of 100K, 500K, 200K, 300K, and 600K (in order), how would each of the First-fit, Best-fit, and Worst-fit algorithms place processes of 212K, 417K, 112K, and 426K (in order)? Which algorithm makes the most efficient use of memory?

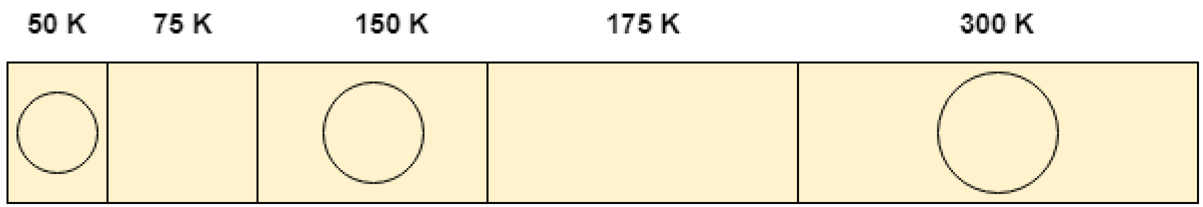
First-Fit:
212K is put in 500K partition.
417K is put in 600K partition.
112K is put in 288K partition (new partition 288K = 500K - 212K).
426K must wait.

Best-Fit:
212K is put in 300K partition.
417K is put in 500K partition.
112K is put in 200K partition.
426K is put in 600K partition.

Worst-Fit:
212K is put in 600K partition.
417K is put in 500K partition.
112K is put in 388K partition.
426K must wait.

In this example, Best-Fit turns out to be the best.

- Q. Process requests are given as; **25 K , 50 K , 100 K , 75 K**



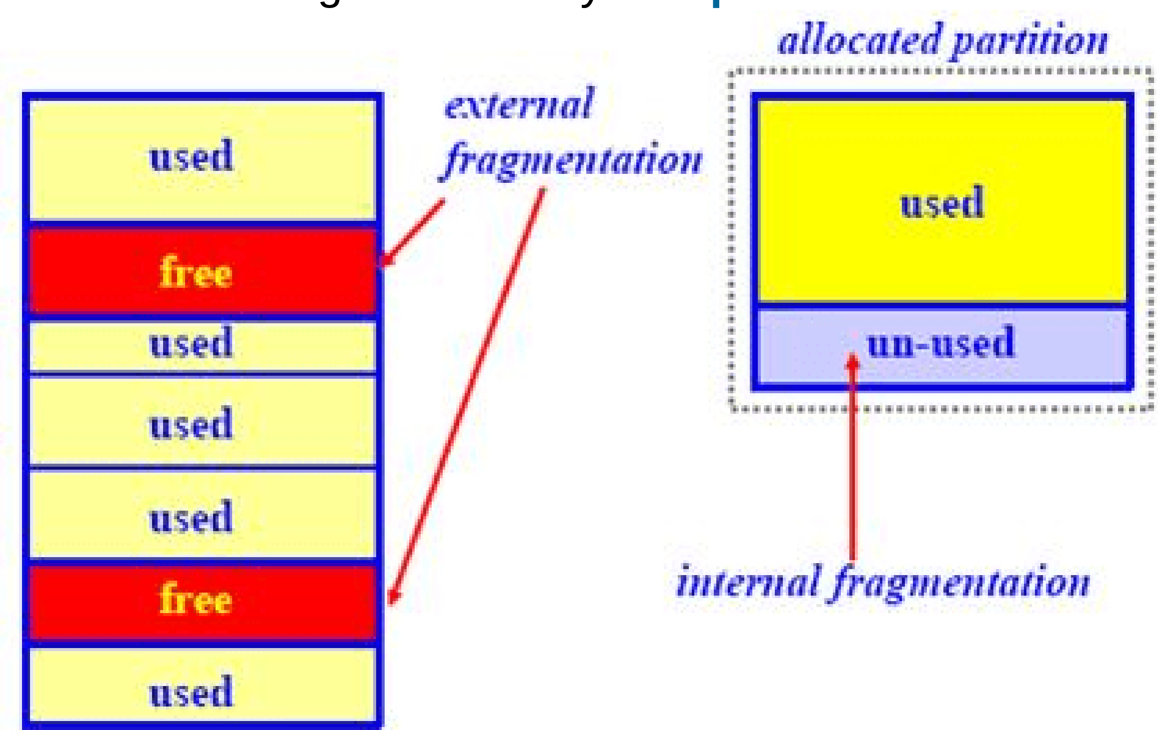
Determine the algorithm which can optimally satisfy this requirement.
1.First Fit algorithm
2.Best Fit Algorithm
3.Neither of the two
4.Both of them

- Consider 6 memory partitions of size 200 KB, 400 KB, 600 KB, 500 KB, 300 K Band 250 KB, where KB refers to kilobyte. These partitions need to be allotted to four processes of sizes 357 KB, 210 KB, 468 KB, 491 KB in that order. If the best-fit algorithm is used, which partitions are NOT allotted to any process?

1.200 KB and 300 KB
2.200 KB and 250 KB
3.250 KB and 300 KB
4.300 KB and 400

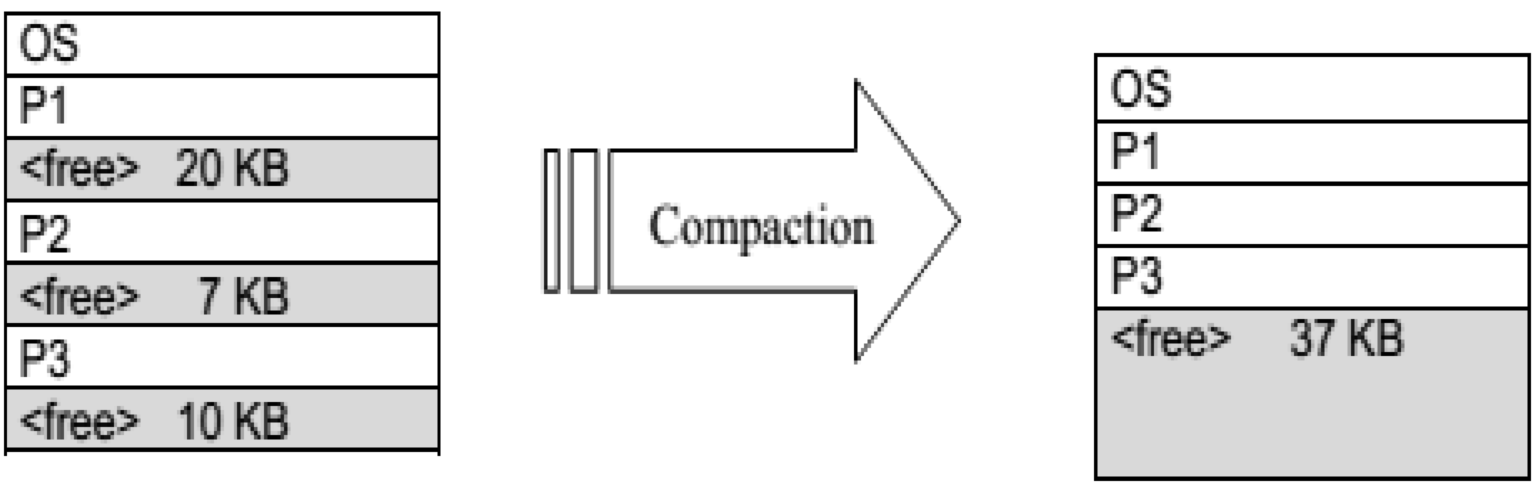
Fragmentation

- **External Fragmentation** – total memory space available to satisfy a request, but it is not contiguous
- **Internal Fragmentation** – allocated memory may be slightly larger than requested memory; this size difference is memory internal to a partition, but not being used
- First fit analysis reveals that given *N* blocks allocated, 0.5 *N* blocks lost to fragmentation
  - 1/2 may be unusable -> **50-percent rule**
- Can reduce external fragmentation by **compaction**



Compaction

- Shuffle memory contents to place all free memory together in one large block
- Compaction is possible *only* if relocation is dynamic, and is done at execution time
- I/O problem. I/O done while compaction. Data arrives in wrong place.
- Solutions to I/O problem
  - Latch process in memory while it is involved in I/O
  - Do I/O only into OS buffers
- Now consider that backing store has same fragmentation problems but on disks



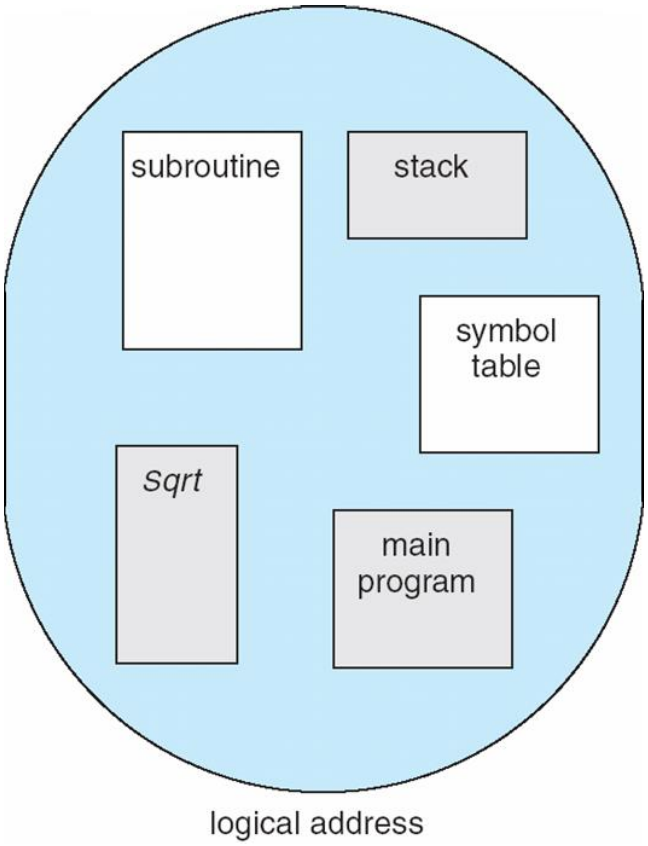
Non-contiguous Allocation

- Allow a process to reside in different locations on memory
  - Segmentation
  - Paging
  - Paged segmentation

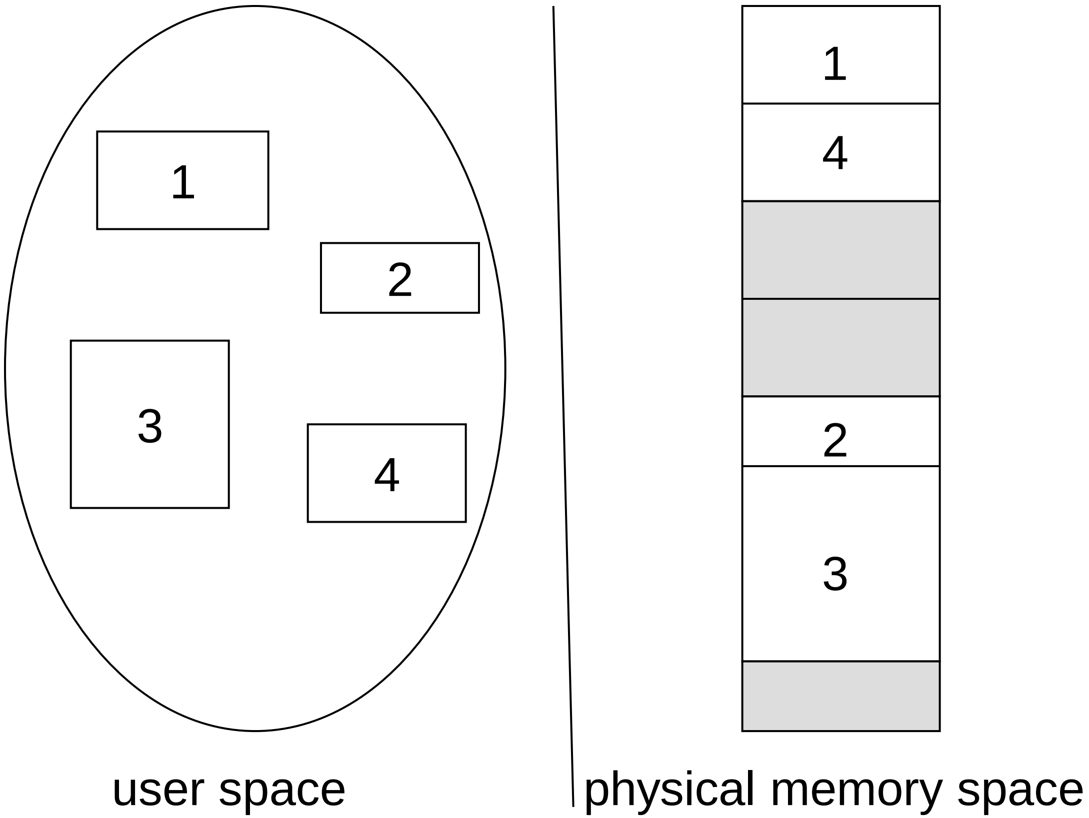
Segmentation

- Memory-management scheme that supports user view of memory
- A program is a collection of segments
- A segment is a logical unit such as:
  - main program
  - procedure
  - function
  - method
  - object
  - local variables, global variables
  - common block
  - stack
  - symbol table
  - arrays

User's View of a Program



Logical View of Segmentation



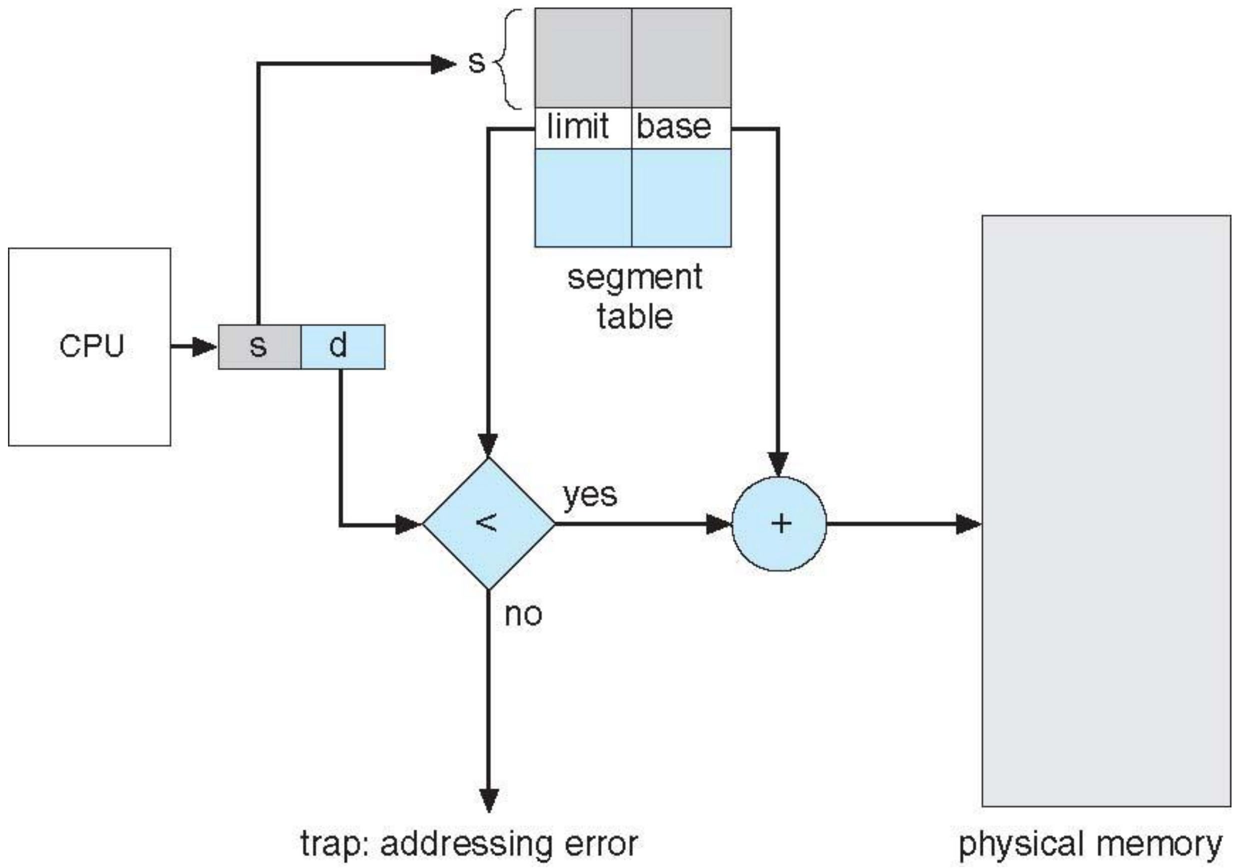
Segmentation Architecture

- Logical address consists of a two tuple:

  $<$segment-number, offset$>$

- **Segment table** – maps two-dimensional physical addresses; each table entry has:
  - **base** – contains the starting physical address where the segments reside in memory
  - **limit** – specifies the length of the segment
- **Segment-table base register** (**STBR**) points to the segment table's location in memory
- **Segment-table length register** (**STLR**) indicates number of segments used by a program
  - Segment number ***s*** is legal if ***s*** < **STLR**

Segmentation Hardware



Segmentation

- Physical address space of a process can be noncontiguous;
- Varying size memory chunks
- External fragmentation

Paging

- Physical address space of a process can be noncontiguous; process is allocated physical memory whenever the latter is available
  - Avoids external fragmentation
  - Avoids problem of varying sized memory chunks
- Divide physical memory into fixed-sized blocks called **frames**
  - Size is power of 2, between 512 bytes and 16 Mbytes
- Divide logical memory into blocks of same size called **pages**
- Keep track of all free frames
- To run a program of size ***N*** pages, need to find ***N*** free frames and load program
- Set up a **page table** to translate logical to physical addresses
- Backing store likewise split into **blocks**
- Still have Internal fragmentation

Address Translation Scheme

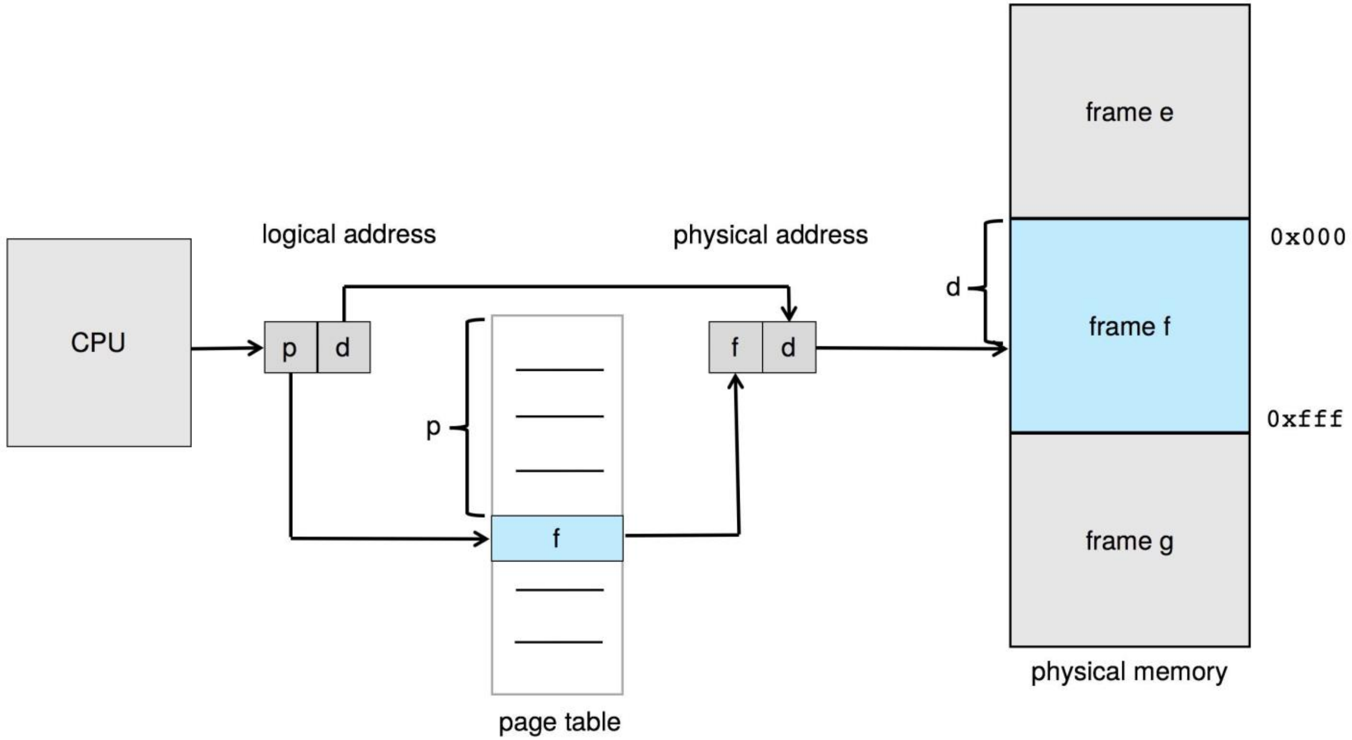
- Address generated by CPU is divided into:
  - **Page number** (***p***) – used as an index into a **page table** which contains base address of each page in physical memory
  - **Page offset** (***d***) – combined with base address to define the physical memory address that is sent to the memory unit

page number	page offset
p	*d*

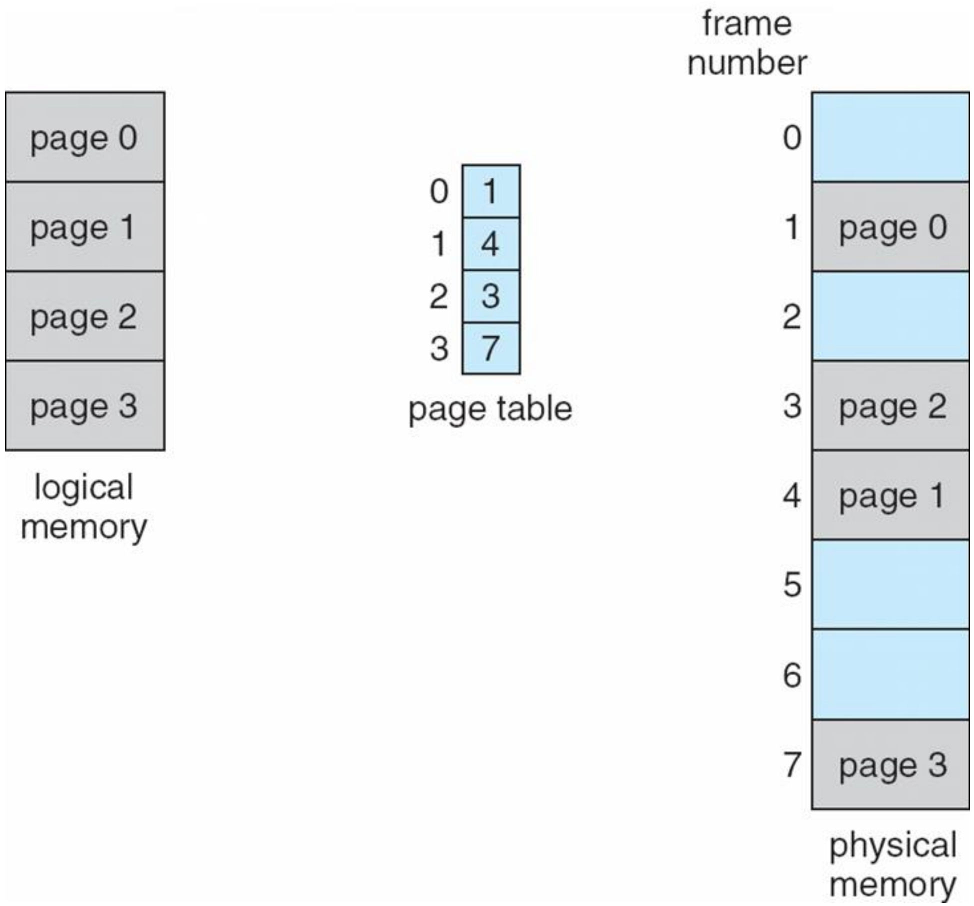
- For given logical address space 2^m and page size 2^n

page number	page offset
p	*d*
$m - n$	n

Paging Hardware

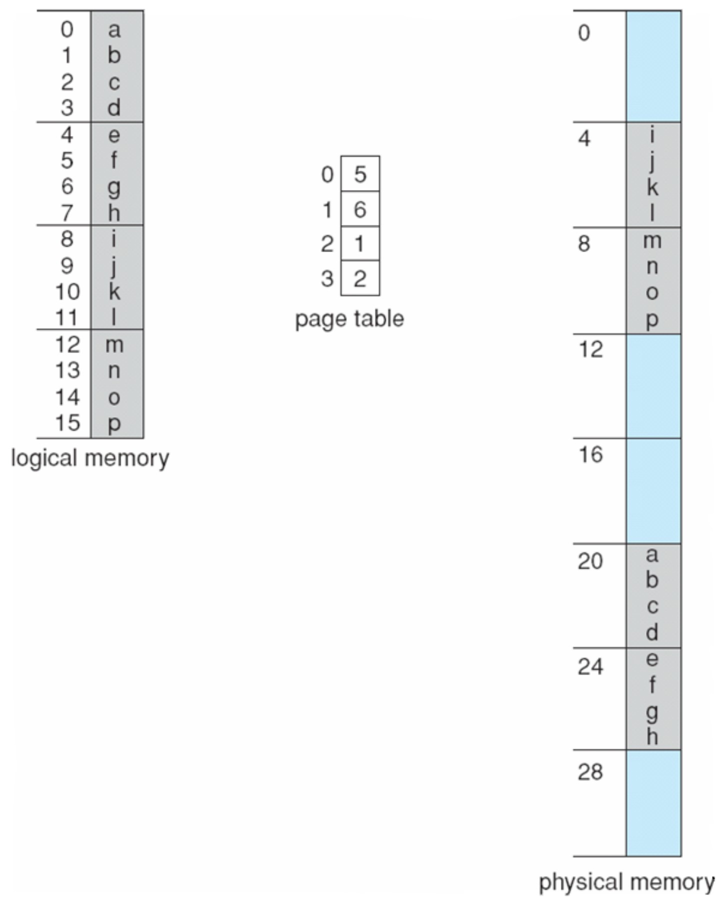


Paging Model of Logical and Physical Memory



Paging Example

- A page size of 4 bytes
- A physical memory of 32 bytes (8 frames)

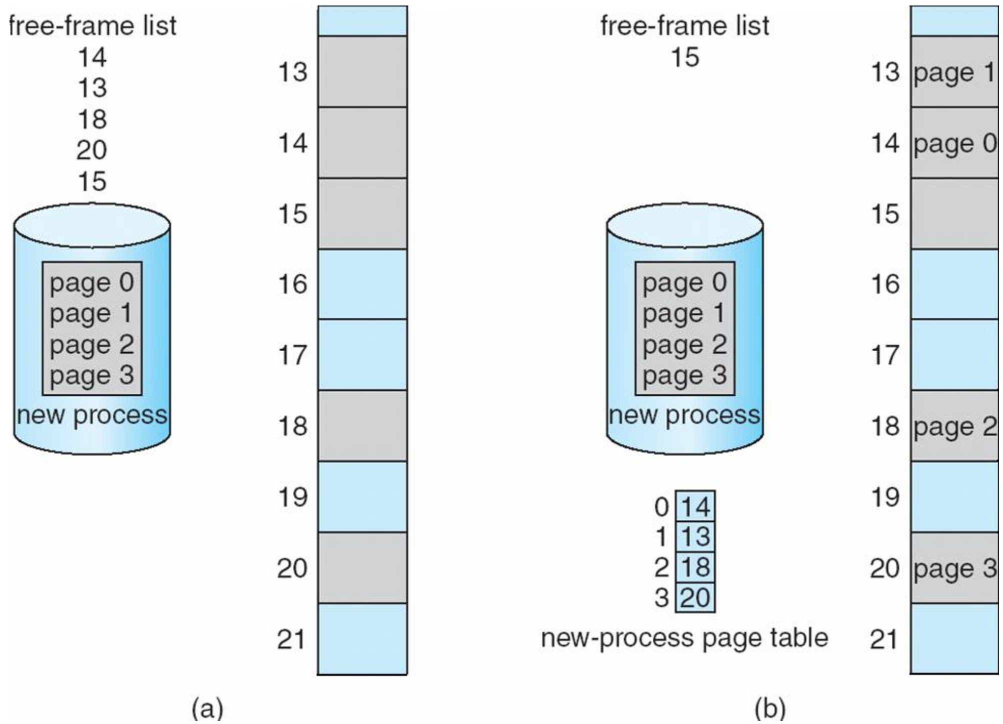


Paging -- Calculating internal fragmentation

- Page size = 2,048 bytes
- Process size = 72,766 bytes
- 35 pages + 1,086 bytes
- Internal fragmentation of 2,048 - 1,086 = 962 bytes
- Worst case fragmentation = 1 frame – 1 byte
- On average fragmentation = 1 / 2 frame size
- So small frame sizes desirable?
- But each page table entry takes memory to track
- Page sizes growing over time
  - Solaris supports two-page sizes – 8 KB and 4 MB

Allocating Free Frames

- Keep a list of free frames
- Example of frame allocation



- New process arrives: (a) before allocation (b) after allocation

Implementation of Page Table

- Page table is kept in main memory
  - **Page-table base register** (**PTBR**) points to the page table
  - **Page-table length register** (**PTLR**) indicates size of the page table
- In this scheme every data/instruction access requires two memory accesses
  - One for the page table and one for the data / instruction
  - Run at half a speed!
- The two-memory access problem can be solved by the use of a special fast-lookup hardware cache called **translation look-aside buffers** (**TLBs**) (also called **associative memory**).

Associative Memory Hardware

- Associative memory – parallel search

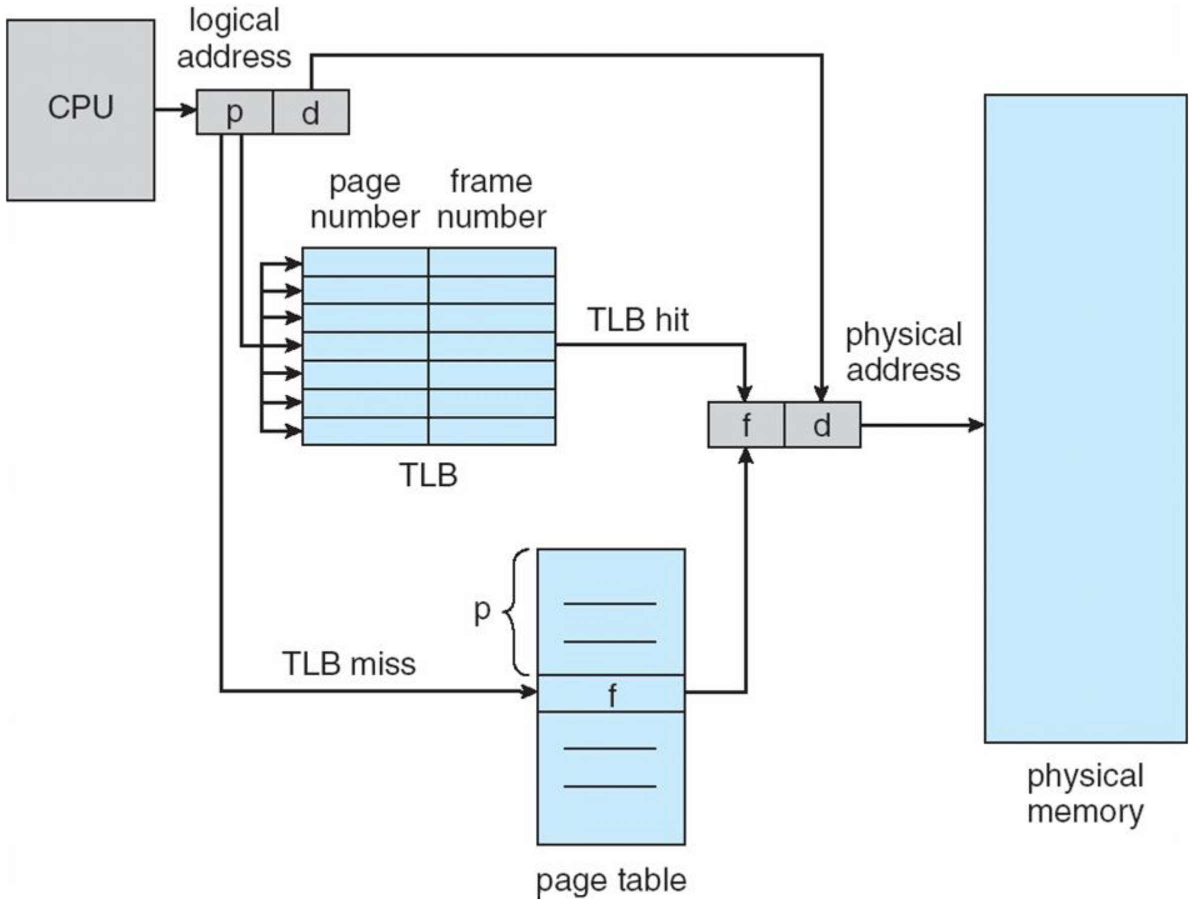
Page #	Frame #

- Address translation (p, d)
  - If p is in associative register, get frame # out
  - Otherwise get frame # from page table in memory

Translation Look-Aside Buffer

- TLBs typically small (64 to 1,024 entries)
- On a TLB miss, value is loaded into the TLB for faster access next time
  - Replacement policies must be considered
  - Some entries can be **wired down** for permanent fast access
- Some TLBs store **address-space identifiers** (**ASIDs**) in each TLB entry – uniquely identifies each process to provide address-space protection for that process
  - Otherwise need to flush the TLB at every context switch

Paging Hardware With TLB



Effective Access Time

- Hit ratio – percentage of times that a page number is found in the TLB
- An 80% hit ratio means that we find the desired page number in the TLB 80% of the time.
- Suppose that it takes 10 nanoseconds to access memory.
  - If we find the desired page in TLB then a mapped-memory access take 10 nanoseconds
  - Otherwise, we need two memory access, so it is 20 nanoseconds
- **Effective Access Time** (**EAT**)

  EAT = 0.80 x 10 + 0.20 x 20 = 12 nanoseconds

  implying 20% slowdown in access time
- Consider a more realistic hit ratio of 99%,

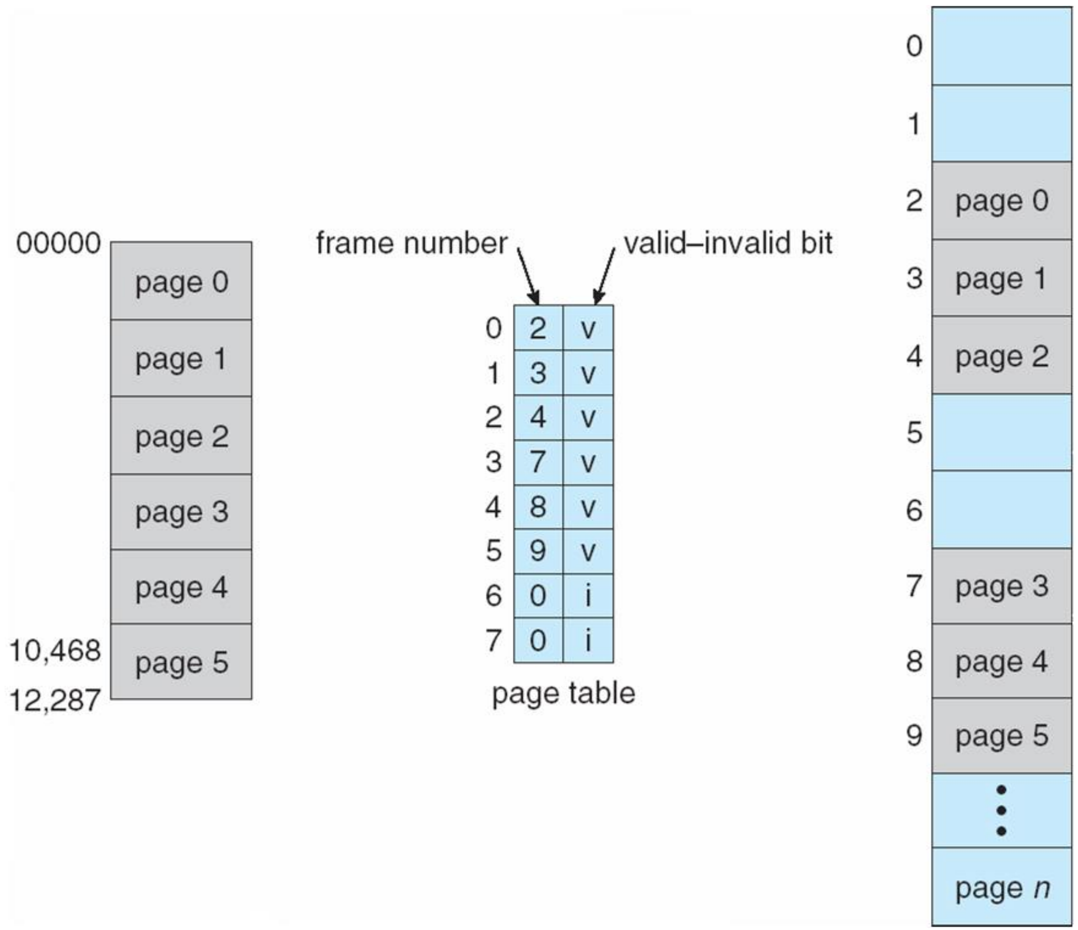
  EAT = 0.99 x 10 + 0.01 x 20 = 10.1 nanoseconds

  implying only 1% slowdown in access time.

Memory Protection

- Memory protection implemented by associating protection bit with each frame to indicate if access is allowed
- **Valid-invalid** bit attached to each entry in the page table:
  - "valid" indicates that the associated page is in the process' logical address space, and is thus a legal page
  - "invalid" indicates that the page is not in the process' logical address space
  - Or use **page-table length register** (**PTLR**)
- Any violations result in a trap to the kernel
- Can also add more bits to indicate if read-only, read-write, execute-only is allowed.

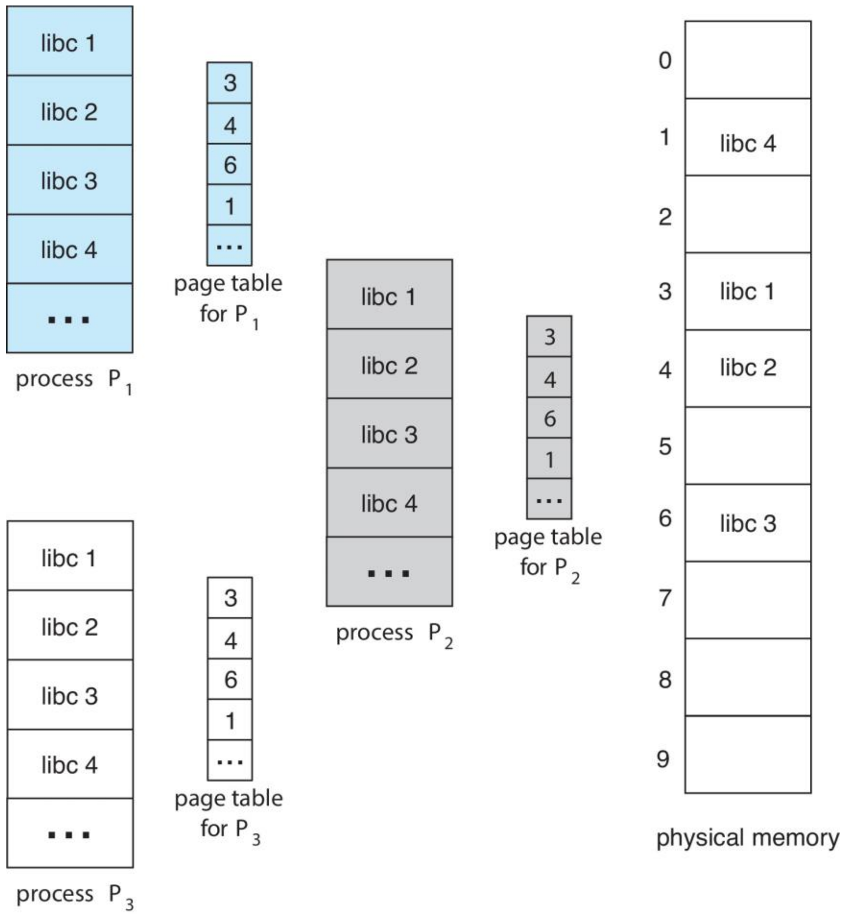
Valid (v) or Invalid (i) Bit In A Page Table



Shared Pages

- **Shared code**
  - One copy of read-only (**reentrant**) code shared among processes (i.e., text editors, compilers, window systems)
  - Similar to multiple threads sharing the same process space
  - Also useful for interprocess communication if sharing of read-write pages is allowed
- **Private code and data**
  - Each process keeps a separate copy of the code and data
  - The pages for the private code and data can appear anywhere in the logical address space

Shared Pages Example

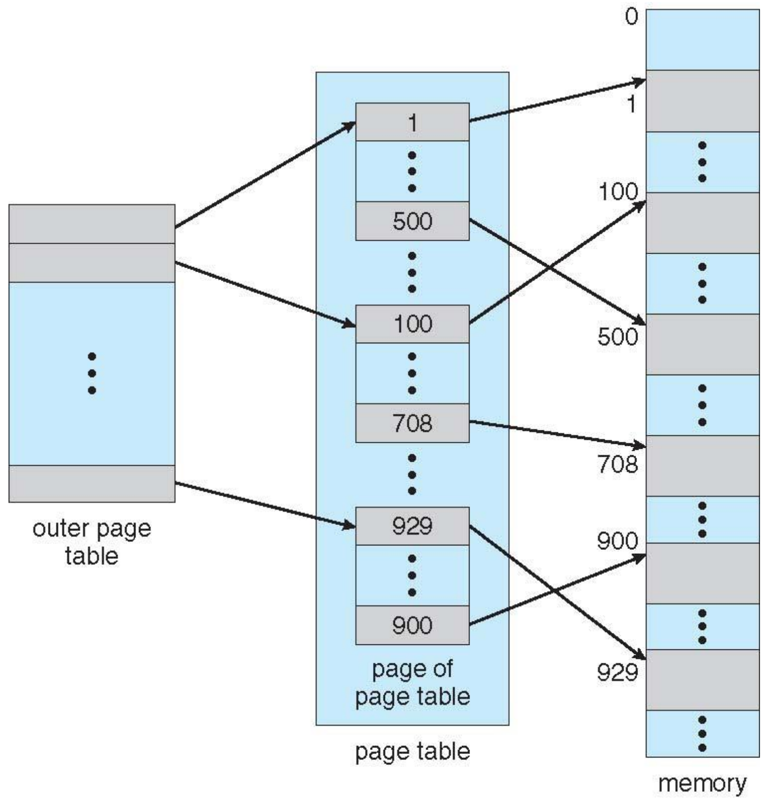


Structure of the Page Table

- Memory structures for paging can get huge using straight-forward methods
  - Consider a 32-bit logical address space as on modern computers
  - Page size of 1 KB (2^{10})
  - Page table would have 1 million entries (2^{32} / 2^{10})
  - If each entry is 4 bytes ➔ each process requires 16 MB of physical address space for the page table alone
    - Do not want to allocate that contiguously in main memory
  - One simple solution is to divide the page table into smaller units
    - Hierarchical Paging
    - Hashed Page Tables
    - Inverted Page Tables

Hierarchical Page Tables

- Break up the logical address space into multiple page tables
- A simple technique is a two-level page table
- We then page the page table



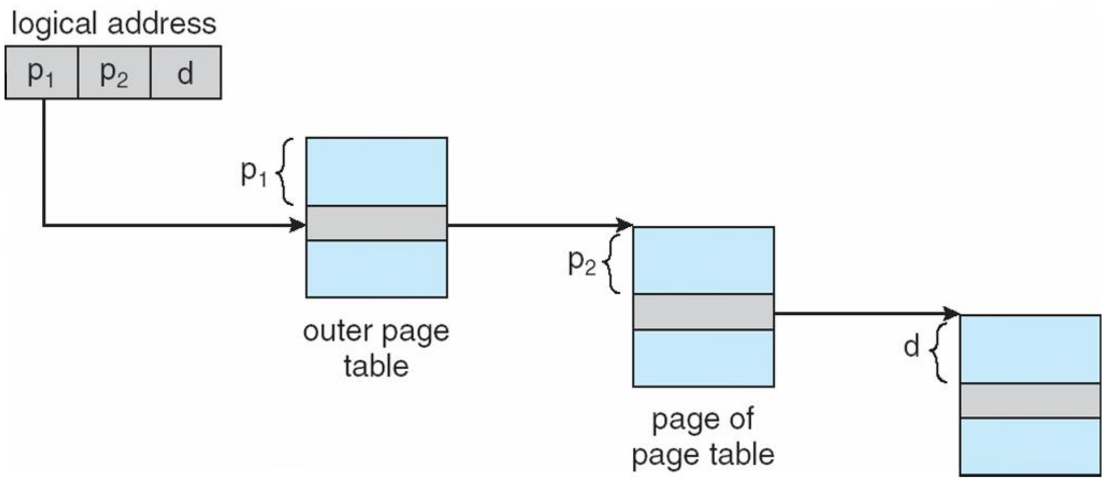
Two-Level Paging Example

- A logical address (on 32-bit machine with 1K page size) is divided into:
  - A page number consisting of 22 bits
  - A page offset consisting of 10 bits
- Since the page table is paged, the page number is further divided into:
  - A 12-bit page number
  - A 10-bit page offset
- Thus, a logical address is as follows:

page number		page offset
p_1	p_2	d
12	10	10

- where p_1 is an index into the outer page table, and p_2 is the displacement within the page of the inner page table
- Known as **forward-mapped page table**

Address-Translation Scheme



64-bit Logical Address Space

- Even two-level paging scheme not sufficient
- If page size is 4 KB (2^{12})
  - Then page table has 2^{52} entries
  - If two level scheme, inner page tables could be 2^{10} 4-byte entries
  - Address would look like

outer page	inner page	offset
p_1	p_2	d
42	10	12

  - Outer page table has 2^{42} entries or 2^{44} bytes
  - One solution is to add a 2nd outer page table
  - But in the following example the 2nd outer page table is still 2^{34} bytes in size
    - And possibly 4 memory access to get to one physical memory location

Three-level Paging Scheme

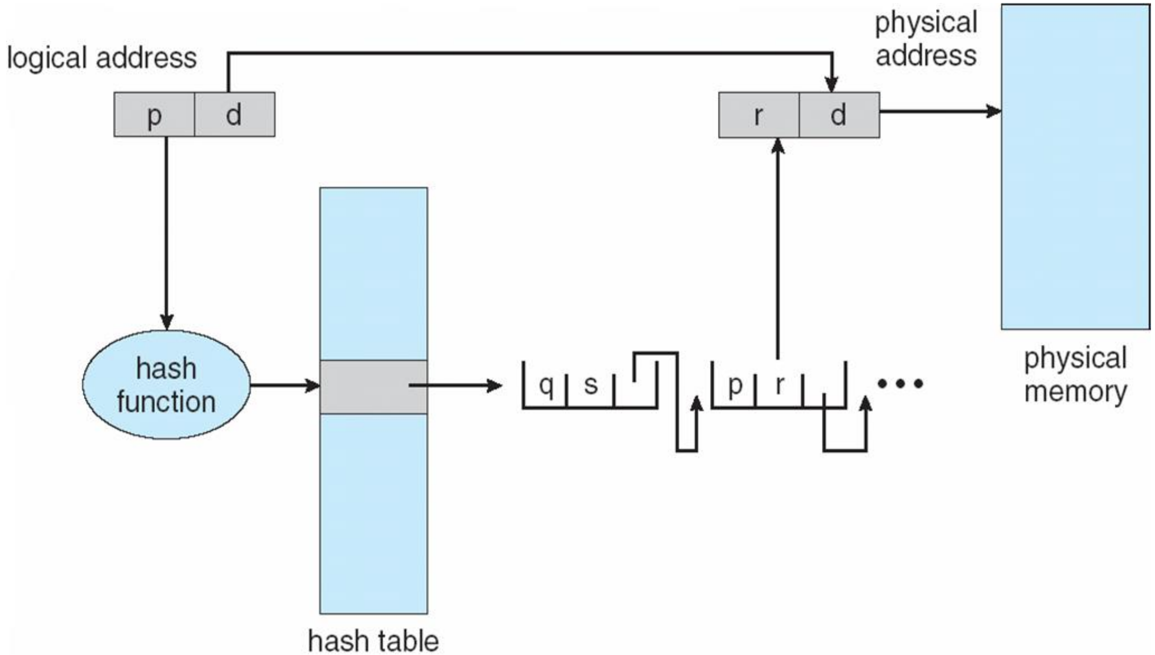
outer page	inner page	offset
p_1	p_2	d
42	10	12

2nd outer page	outer page	inner page	offset
p_1	p_2	p_3	d
32	10	10	12

Hashed Page Tables

- Used in architecture with address spaces > 32 bits
- The virtual page number is hashed into a page table
  - This page table contains a chain of elements hashing to the same location
- Each element contains
  1. The virtual page number
  2. The value of the mapped page frame
  3. A pointer to the next element
- Virtual page numbers are compared in this chain searching for a match
  - If a match is found, the corresponding physical frame is extracted

Hashed Page Table Hardware

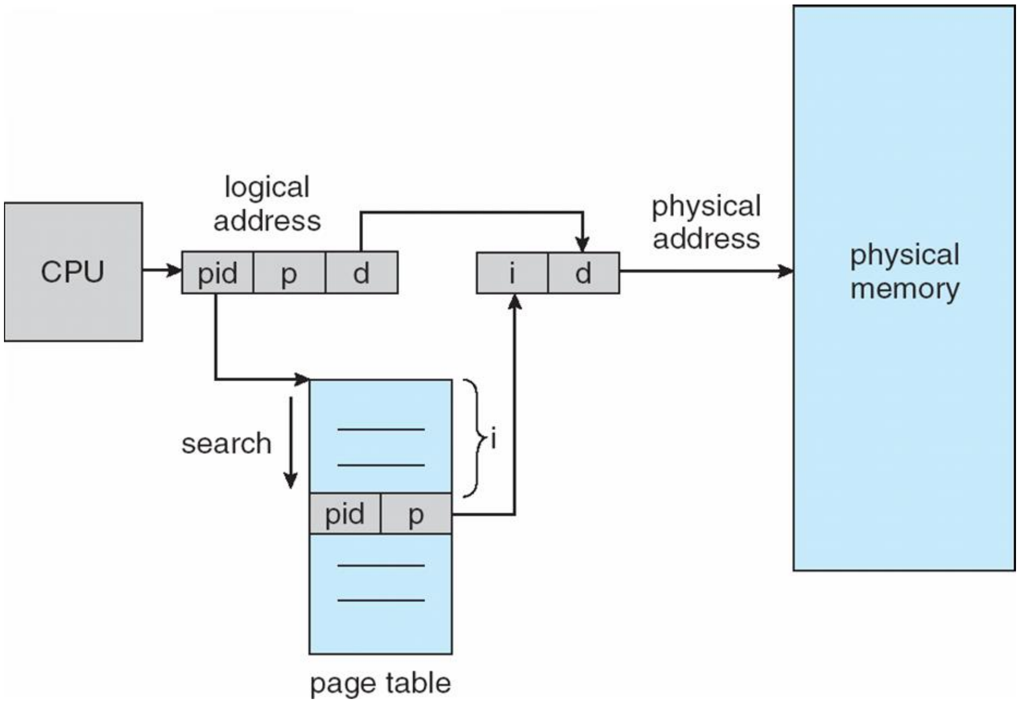


Clustered Hashed Page Tables

- Variation for 64-bit addresses
- Similar to hashed but each entry refers to several pages (such as 16) rather than 1
- Especially useful for **sparse** address spaces (where memory references are non-contiguous and scattered)

Inverted Page Table

- Rather than having each process keep a page table and track of all possible logical pages, track all physical pages
- One entry for each real page of memory
- Entry consists of the virtual address of the page stored in that real memory location, with information about the process that owns that page



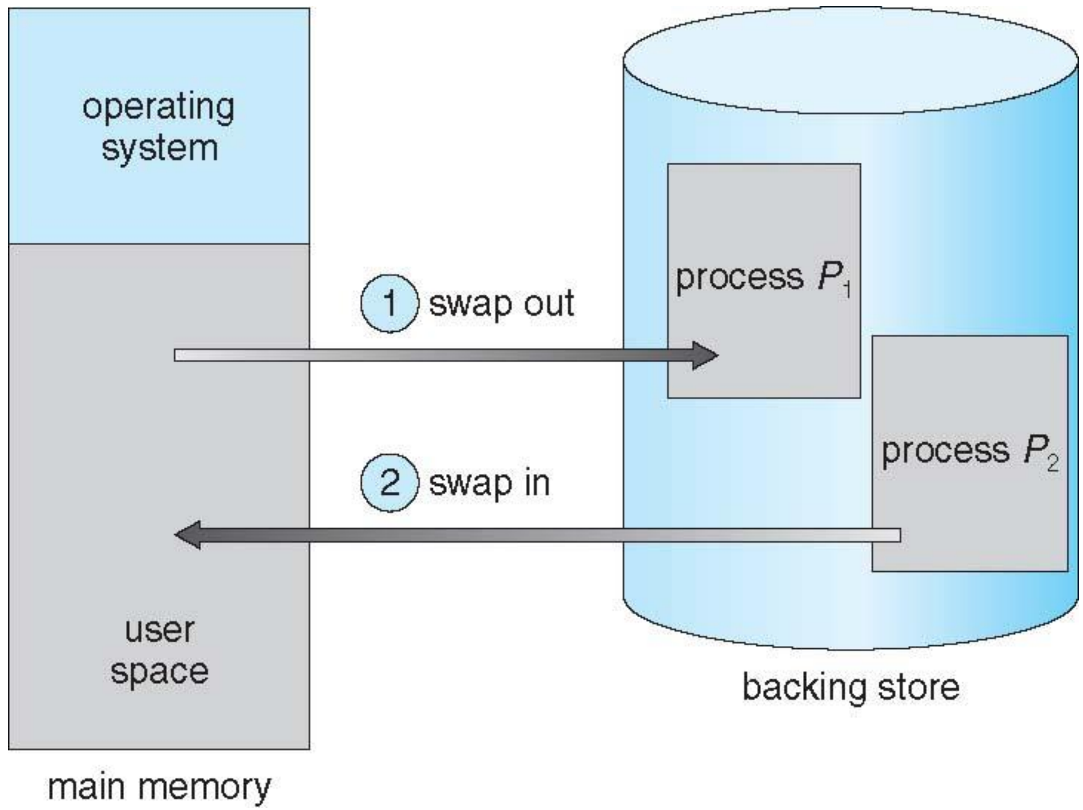
Inverted Page Table (Cont.)

- Decreases memory needed to store each page table, but increases time needed to search the table when a page reference occurs
- Use hash table to limit the search to one (or at most a few) page-table entries
  - TLB can accelerate access
- But how to implement shared memory?
  - One mapping of a virtual address to the shared physical address

Swapping

- A process can be **swapped** temporarily out of memory to a backing store, and then brought **back** into memory for continued execution
- **Backing store** – fast disk large enough to accommodate copies of all memory images for all users; must provide direct access to these memory images
- **Roll out, roll in** – swapping variant used for priority-based scheduling algorithms; lower-priority process is swapped out so higher-priority process can be loaded and executed
- Major part of swap time is transfer time; total transfer time is directly proportional to the amount of memory swapped
- System maintains a **ready queue** of ready-to-run processes which have memory images on disk

Schematic View of Swapping



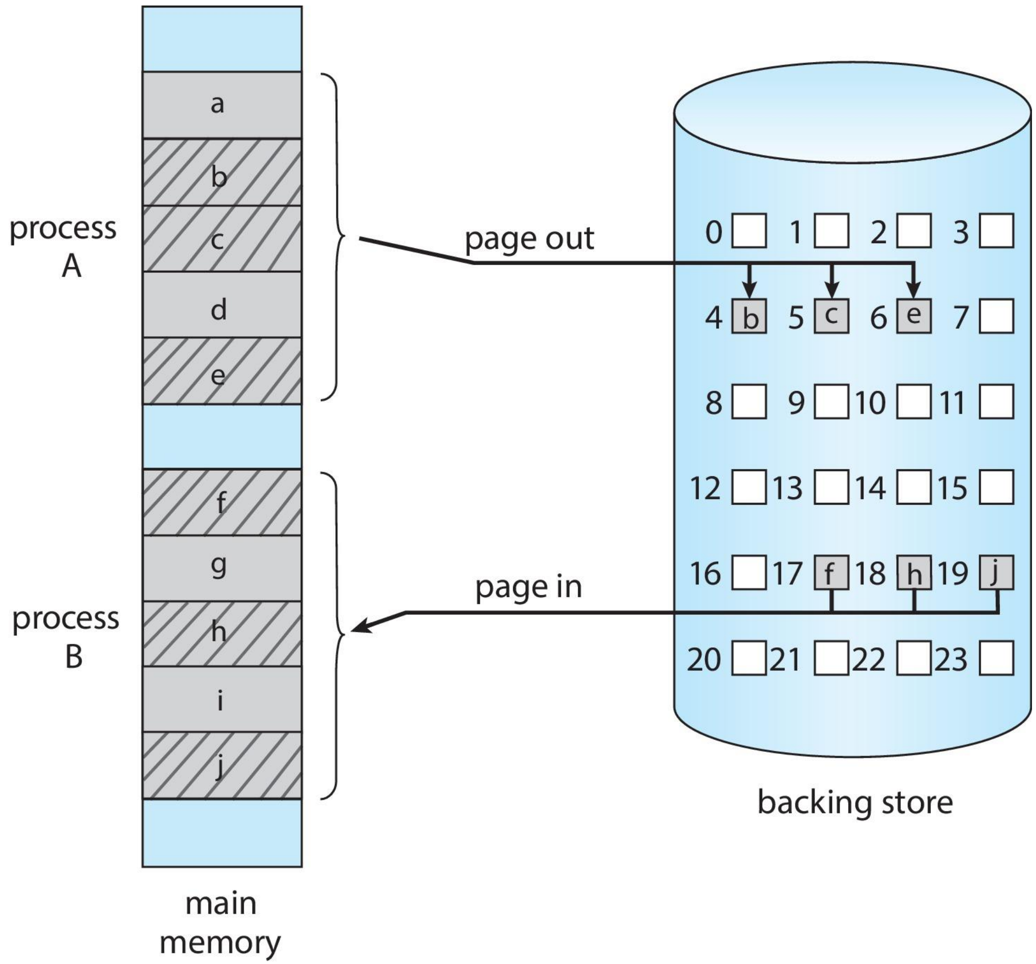
Swapping (Cont.)

- Does the swapped-out process need to swap back-in to same physical addresses?
  - Depends on address binding method
- Modified versions of swapping are found on many systems (i.e., UNIX, Linux, and Windows)
  - Swapping normally disabled
  - Started if more than threshold amount of memory allocated
  - Disabled again once memory demand reduced below threshold
- Other constraints on swapping
  - Pending I/O – can't swap out as I/O would occur to wrong process
  - Or always transfer I/O to kernel space, then to I/O device
    - Known as **double buffering**, adds overhead

Context Switch Time including Swapping

- If next processes to be obtain CPU is not in memory, need to swap out a process and swap in target process
- Context switch time can then be very high
- 100MB process swapping to hard disk with transfer rate of 50MB/sec
  - Swap out time of 2000 milliseconds
  - Plus swap in of same sized process
  - Total context switch swapping component time of 4000 milliseconds (4 seconds)

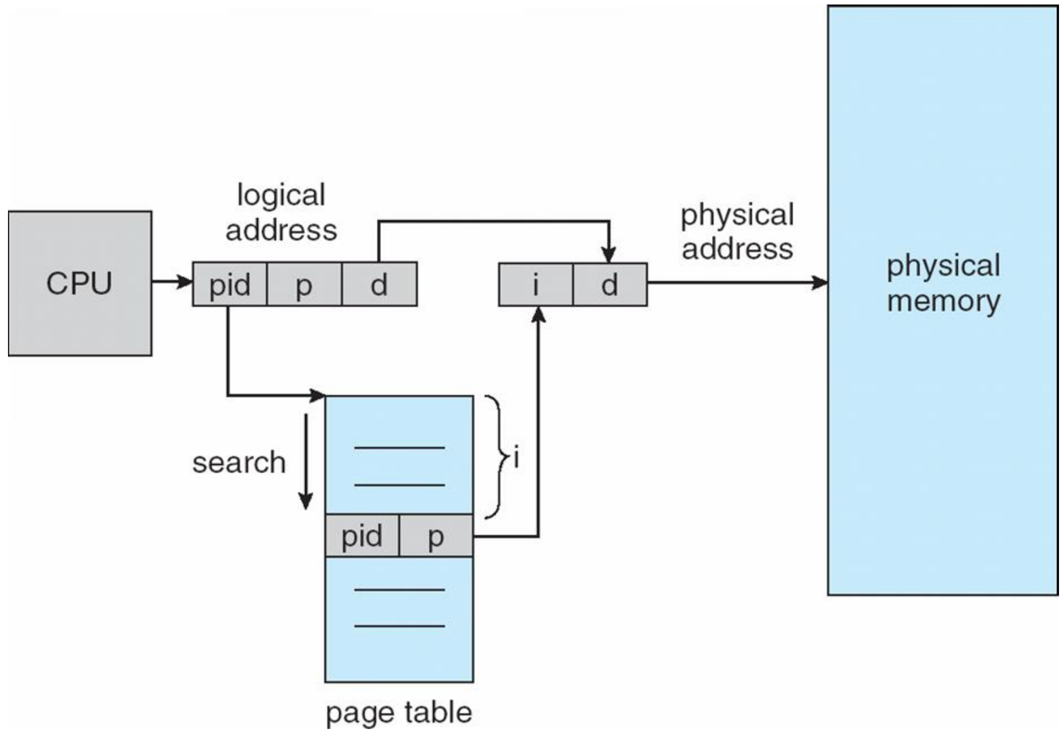
Swapping with Paging



Swapping on Mobile Systems

- Not typically supported -- Flash memory based
  - Small amount of space
  - Limited number of write cycles
  - Poor throughput between flash memory and CPU on mobile platform
- Instead use other methods to free memory if low
  - iOS ***asks*** apps to voluntarily relinquish allocated memory
    - Read-only data thrown out and reloaded from flash if needed
    - Failure to free can result in termination
  - Android terminates apps if low free memory, but first writes **application state** to flash for fast restart
  - Both OSes support paging as discussed below

Inverted Page Table Architecture



Swapping (Cont.)

- Other constraints on swapping
  - Pending I/O – can't swap out as I/O would occur to wrong process
  - Or always transfer I/O to kernel space, then to I/O device
    - Known as **double buffering**, adds overhead
- Standard swapping not used in modern operating systems
  - But modified version common
    - Swap only when free memory extremely low